

Name: Tridib Banik

Student# 400514461

The two (non-primary key) indexes that I would recommend are:

1. **Composite index on Complaints(Category, Timestamp, TripID):** The properties of the index would be B-tree type, unclustered, composite. Many queries filter complaints by category and timestamp ranges (e.g., “driver conduct” in Q2 or “overcrowding” in Q7) and then join on TripID to link to trip or driver information. A composite index on (Category, Timestamp, TripID) allows efficient filtering on category and timestamp while supporting fast lookups for the associated trip records. I think this index will help Q2 and Q7 queries, which both rely on complaint filtering and trip joins. Finally, the use of this index reduces full table scans on Complaints when processing time-based or category-based lookups.
2. **Composite index on Takes(TripID, Date):** The properties of the index would be B-tree type, unclustered, composite. Queries frequently join Takes with Trips and filter or group by date ranges (e.g., riders who took trips in specific periods in Q3, riders who didn’t travel in December in Q5, and passenger counts per trip in Q8). Indexing (TripID, Date) speeds up joins and improves performance for range-based filters on the Takes table. I think this index will help Q3, Q5, and Q8 queries by minimizing the number of scanned rows and enabling efficient join operations. Finally, the use of this index avoids sequential scans over large Takes datasets and improves the query planner’s ability to use range and equality conditions together.